

5.3 Construction of DC Machines

A dc machine consists of the following parts

1. Yoke or Frame
2. Field system
3. Armature
4. Commutator
5. Brushes
6. Bearings

1. **Yoke:** It is the outer frame of the machine. Frame serves the following purposes

- (a) It acts as protective cover for the machines
- (b) It houses field system and supports the armature through bearings
- (c) It carries the working flux.

The material of yoke should have high permeability and high mechanical strength. Generally, yoke is made of cast steel.

2. **Field System:** It is basically a system of electromagnets which is excited by dc supply. The field system produces the working flux of the machine. It is the stationary part of the machine and is bolted to the frame of the machine. For large dc machine, field system consists of pole core, pole shoes and pole (Field) winding as shown in fig. 5.6.

(a) **Pole Core:** The pole core carries, field winding. It is built using thin steel laminations. One end of the pole core is bolted to the frame and other end has pole shoe.

(b) **Pole Shoes:** The pole shoe is a projected part of the pole core and has large crosssectional area. It serves following purposes.

- (i) It spread the flux uniformly in the air gap and offers low reluctance path to the flux.
- (ii) It supports field winding.

(c) **Field Coils:** Field coils are made of copper wire or copper strip. They are former wound and inserted around the pole core. When field windings are excited by dc supply, they become electromagnets and produce the working flux.

3. **Armature:** Armature is a system of conductors or coil which is free to rotate on the supported bearings. Emfs and torques are developed in the armature coils. The armature consists of the following parts.

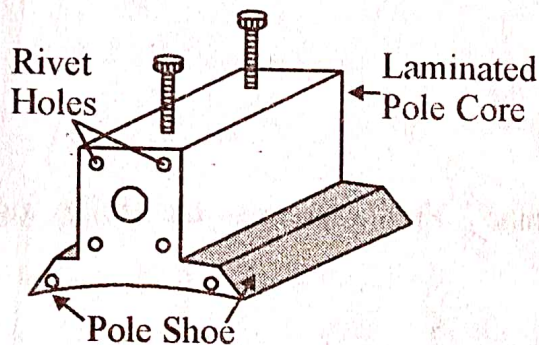


Fig. 5.6 Pole core and pole shoes

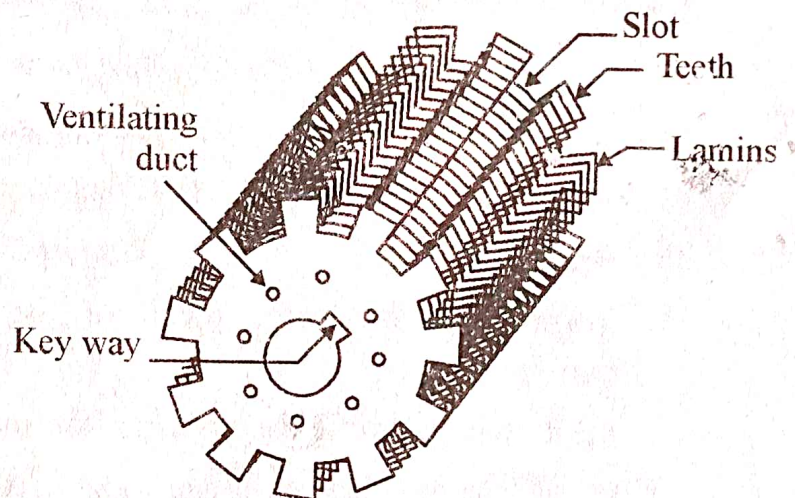


Fig. 5.7 Armature core

- (a) **Armature Core** : The armature core is a solid cylindrical structure. It is made of high permeability thin silicon content steel laminations. The outer periphery of the core has slots to carry armature winding as shown in fig. 5.7.
- (b) **Armature Winding** : The armature winding (conductors) is generally made of copper wires. The winding is generally former wound. The winding may be single layer (one conductor in each slot) or multilayer winding. Depending upon the end connections to the armature conductors, winding may be classified in various types such as lap winding, wave winding etc. The type of winding decides the voltage and current rating of the machine.

If Z is the number of conductors and P is the number of poles

No of parallel path

$$A = P \quad (\text{for lap winding})$$

$$A = 2 \quad (\text{for wave winding})$$

No of conductor in series in each parallel path

$$= \frac{Z}{P} \quad (\text{for lap winding})$$

$$= \frac{Z}{2} \quad (\text{for wave winding})$$

4. **Commutator** : The commutator is the important part of the dc machine. It performs the following functions

- (a) To collect (or to allow) the current from (or to) the armature conductor.
- (b) To convert the alternating current of the armature into unidirectional current in the external circuit with the help of brushes and vice-versa.

It is important to note that, the current flowing in the individual armature conductor is alternating in nature.

The commutator is cylindrical in shape and is made of copper. The outer periphery of the commutator has V-shaped grooves or slots to carry commutator segments. The commutator segments are copper bars inserted in the grooves of the commutator. These segments are insulated from each other by mica material. Each commutator segment is connected to two armature conductors with the help of copper lugs or risers. The commutator is mounted on the shaft of the machine on one side of the armature.

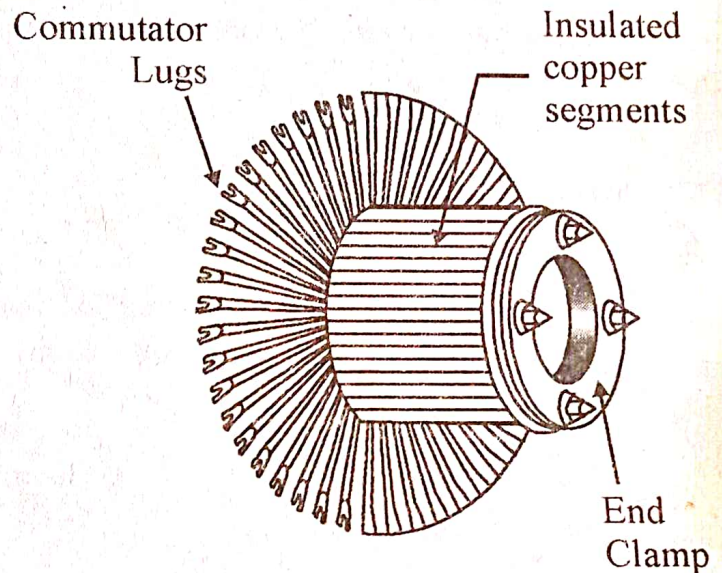


Fig. 5.8 Commutator

5. Brushes : The function of brushes is to collect current from moving commutator.

They are usually made of carbon or graphite. They are housed in brush holds and are in contact with commutator surface with the help of spring pressure.

6. Bearings : The function of bearings is to support the machine shaft with minimum friction. For dc machines, generally, ball bearing or roller bearings are used.

Besides the above mentioned parts, large dc machines also have compoles, compensating windings, cooling fans etc.

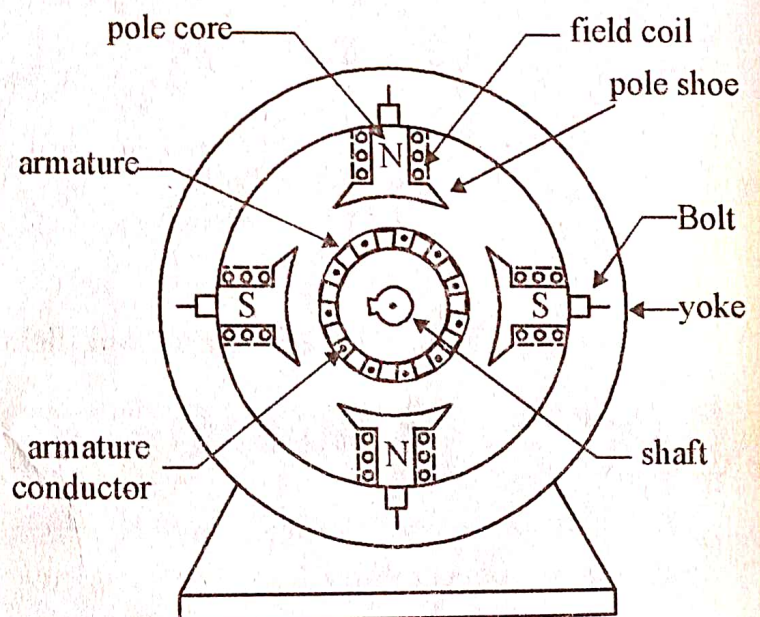


Fig. 5.9 Overall view of dc machine

5.4 EMF Equation of DC Generator

DC generators are classified on the basis of the excitation of field winding. The field winding produces the required working flux. This flux induces emf in the armature circuit by generator action. The emf induced in a practical generator is given by the expression

$$E = \phi Z n \left(\frac{P}{A} \right) \text{ Volts} \quad \dots(5.7)$$

where,

ϕ = flux/pole (wb)

Z = Total number of conductor in armature

n = speed of armature in rps

P = Number of poles

A = Number of parallel paths in the armature

$A = 2$ (for wave winding)

$A = P$ (for lap winding)

The electrical equivalent circuit of armature and field system of a dc generator is as shown in fig. 5.10. Here, V is the terminal voltage, I_a is the armature current and r_a is the resistance of armature circuit. The terminal voltage of dc generator is given by

$$V = E - I_a r_a$$

Where I_a is armature current and r_a is armature resistance

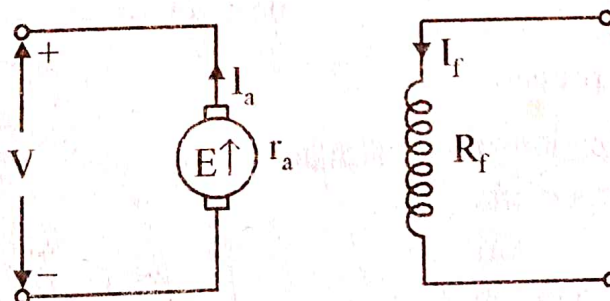


Fig. 5.10 Armature and field equivalent circuits